Yu SHUAI

Department of Mechanical and Aerospace Engineering, Princeton University
yu_shuai@princeton.edu | Personal website: yu-shuai-pu.github.io

RESEARCH INTERESTS

Fluid dynamics; flow control; model reduction; data-driven machine learning; uncertainty quantification.

EDUCATION

Princeton University | Princeton, NJ, USA Aug 2022–Present
Ph.D. candidate in Mechanical and Aerospace Engineering
Advisor: Clarence W. Rowley | Expected defense: July 2027

Peking University | Beijing, China Sep 2018–Jul 2022
B.Sc. in Theoretical and Applied Mechanics

PUBLICATIONS

Published

1. Y. Shuai, C. Liu, and D. F. Gayme. "Structured input–output analysis of oblique laminar–turbulent patterns in plane Couette–Poiseuille flow." *International Journal of Heat and Fluid Flow*, 103, 109207 (2023). doi: 10.1016/j.ijheatfluidflow.2023.109207
2. C. Liu, **Y. Shuai**, A. Rath, and D. F. Gayme. "A structured input-output approach to characterizing optimal perturbations in wall-bounded shear flows." *2023 American Control Conference (ACC)*, pp. 2319–2325 (2023). doi: 10.23919/ACC55779.2023.10156058
3. Y. Shuai, C. Liu, and D. F. Gayme. "Structured input-output analysis of oblique turbulent bands in transitional plane Couette-Poiseuille flow." *Proceedings of the 12th International Symposium on Turbulence and Shear Flow Phenomena (TSFP12)*, Osaka, Japan (2022). Proceedings PDF

Submitted

4. Y. Shuai and C. W. Rowley. "Symmetry-reduced model reduction of shift-equivariant systems via operator inference." *Advances in Computational Mathematics* (submitted; in revision). arXiv:2507.18780 (2025).
5. Y. Shuai and C. W. Rowley. "S-R NiTROM: Symmetry-Reduced Non-intrusive Trajectory-based Optimization of ROMs." In preparation.

RESEARCH EXPERIENCE

Doctoral Researcher | Princeton University Aug 2022–Present
Mechanical and Aerospace Engineering, Princeton, NJ, USA | Advisor: Clarence W. Rowley

- **Symmetry-reduced operator inference.** Developed reduced models that learn the shape dynamics of spatially shifting solutions in a co-moving frame and reconstruct the shift amount; validated on the Kuramoto–Sivashinsky equation.
- **Coupled subspace–dynamics learning.** Developed S-R NiTROM (Symmetry-Reduced Non-intrusive Trajectory-based Optimization of ROMs), which learns the projection subspaces and the symmetry-reduced dynamics together, data-driven and non-intrusive, without knowledge of the full-order operators. On the non-reciprocal Cahn–Hilliard equation it improves on the SR-Galerkin and SR-OpInf benchmarks and generalizes more reliably than other current trajectory-optimized models; the learned subspaces explore more directions than the benchmarks do. On a three-dimensional linearized Navier–Stokes example it matches intrusive balanced POD at equal subspace dimension on unseen test data.
- **Lagrangian coherent structures from sparse tracer data** (with Nicholas Conlin and Marcus Hultmark). Proposed a pipeline for detecting Lagrangian coherent structures in the atmospheric surface layer from sparse tracer trajectories: a large-eddy-simulation baseline, 4D-Var estimation and calibration against the measured trajectories, then geometric LCS detection. A baseline velocity-field reconstruction by nearest-neighbour interpolation is in place; the large-eddy simulation and the 4D-Var step are in progress.

Undergraduate Researcher | Peking University Oct 2021–Jun 2022
School of Mathematical Sciences, Beijing, China | Advisor: Zhifei Zhang
Undergraduate thesis. Studied transient growth and transition thresholds for Couette–Poiseuille flow using asymptotic analysis and Chebyshev-collocation eigenvalue computations.

Visiting Student Researcher | Johns Hopkins University May 2021–Oct 2021
Mechanical Engineering, Baltimore, MD, USA | Advisor: Dennice F. Gayme
Applied structured input–output analysis to transitional Couette–Poiseuille flow, characterizing oblique turbulent bands, dominant wavelengths and Reynolds-number scaling.

Undergraduate Project Lead | Peking University Sep 2020–Nov 2021
State Key Laboratory for Turbulence and Complex Systems, Beijing, China | Advisor: Yipeng Shi
Developed wavelet-based, self-similar reconstruction of corrupted rotating-turbulence data and compared it with gappy POD and GAN baselines.

RESEARCH SOFTWARE

SROpInf — Python package for symmetry-reduced operator inference and POD–Galerkin models, with reproducible Kuramoto–Sivashinsky examples and tests.

SELECTED PRESENTATIONS

79th APS Division of Fluid Dynamics Meeting | Orlando, FL, USA Nov 22–24, 2026
Scheduled oral presentation: "Inferring Lagrangian Coherent Structures in the Atmospheric Surface Layer from Sparse Particle Measurements."

SIAM Conference on Mathematics of Data Science (MDS26) | Salt Lake City, UT, USA Nov 16–20, 2026
Scheduled. Minisymposium co-organizer (with Clarence W. Rowley) and oral presentation: "Non-Intrusive Optimization of Reduced-Order Models for Shift-Equivariant Systems based on Symmetry Reduction" in the minisymposium *Data-Driven Nonlinear Reduced-Order Models for Transport-Dominated Systems*.

78th APS Division of Fluid Dynamics Meeting | Houston, TX, USA Nov 23–25, 2025
Oral presentation: "Non-Intrusive Model Reduction of Shift-Equivariant Systems via Data-Driven Projection and Symmetry-Reduced Dynamics."

SIAM Conference on Applications of Dynamical Systems | Denver, CO, USA May 11–15, 2025
Minisymposium co-organizer (with Clarence W. Rowley) and oral presentation: "Reduced-Order Modeling of Shift-Equivariant Systems Using Operator Inference" in the minisymposium *Reduced-Order Modeling of Fluid Flow with Coherent Structures*.

77th APS Division of Fluid Dynamics Meeting | Salt Lake City, UT, USA Nov 24–26, 2024
Oral presentation: "Neural Network-Based Closure Model of the Ensemble-Averaging Dynamics of Turbulent Puffs in Transitional Pipe Flow."

75th APS Division of Fluid Dynamics Meeting | Indianapolis, IN, USA Nov 20–22, 2022
Oral presentation: "Structured Input-Output Analysis of Oblique Laminar-Turbulent Flow Patterns in Transitional Plane Couette-Poiseuille Flow."

TEACHING

Assistant in Instruction, Princeton University:

- **MAE 305: Mathematics in Engineering I** Fall 2026
- **MAE 222: Mechanics of Fluids** Spring 2026
- **MAE 305: Mathematics in Engineering I** Fall 2025
- **MAE 553: Turbulent Flow** Spring 2025
- **MAE 433: Automatic Control Systems** Fall 2024

HONORS AND AWARDS

Guggenheim Second Year Fellowship, Princeton University 2023
Award for Excellent Undergraduate Research Program, Peking University 2021
Huatai Financial Technology Scholarship, Peking University 2021
Boeing Scholarship, Peking University 2020

TECHNICAL SKILLS

Programming Python (NumPy, SciPy, JAX, PyTorch), MATLAB, Fortran, C++
CFD libraries Dedalus, Channelflow 2.0, Openpipeflow
High-performance computing MPI, Slurm